

CSS accent-color Complete Guide

CSS Educational Series

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Contents

1	Introduction to accent-color	2
1.1	What is accent-color?	2
1.1.1	Key Facts	2
1.1.2	Supported Elements	2
1.1.3	Browser Support	2
2	Basic Syntax and Values	4
2.1	Fundamental Syntax	4
2.2	Color Value Types	4
2.3	Inheritance Behavior	6
3	Practical Applications	7
3.1	Form Styling with Brand Colors	7
3.2	Checkbox Styling	10
3.3	Radio Button Styling	12
4	Range and Progress Elements	14
4.1	Range Input Styling	14
4.2	Progress Bar Styling	16
5	Advanced Applications	19
5.1	Multi-Theme Systems	19
5.2	Dynamic Accent Colors	21
6	Fallbacks and Browser Support	23
6.1	Feature Detection	23
6.2	Progressive Enhancement	23
6.3	Browser Compatibility Table	24
7	Color Input with accent-color	25
7.1	Color Input Styling	25
8	Real-World Examples	27
8.1	Settings Panel	27
8.2	Survey Form	30

9 Quick Reference	34
9.1 Complete Property Reference	34
9.2 Best Practices	34
9.3 Supported Elements Summary	35
10 Conclusion	36
10.1 Summary	36
10.2 Key Advantages	36
10.3 Related CSS Features	36

Chapter 1

Introduction to accent-color

1.1 What is accent-color?

The CSS `accent-color` property sets the accent color for certain user-interface controls such as checkboxes, radio buttons, range sliders, progress bars, and color inputs. This property provides a simple way to theme form elements with a brand color without complex customization.

1.1.1 Key Facts

- Controls accent color for form elements (checkboxes, radios, ranges, progress)
- Applies to `<input type="checkbox">`, `<input type="radio">`, `<input type="range">`
- Works with `<progress>` and `<meter>` elements
- Accepts any valid CSS color value
- Respects system color preferences when set to `auto`
- Modern browser support: Chrome 93+, Firefox 92+, Safari 15.4+
- No JavaScript needed for basic theming

1.1.2 Supported Elements

Element	Effect
<code><input type="checkbox"></code>	Checkmark and box color
<code><input type="radio"></code>	Radio button dot color
<code><input type="range"></code>	Slider thumb color
<code><progress></code>	Progress bar fill color
<code><meter></code>	Meter bar color
<code><input type="color"></code>	Color input controls

1.1.3 Browser Support

Browser	Chrome	Firefox	Safari	Edge
accent-color	93+	92+	15.4+	93+

Chapter 2

Basic Syntax and Values

2.1 Fundamental Syntax

```
1  /* Set accent color to a specific color */
2  input[type="checkbox"] {
3      accent-color: #ff6b6b;
4  }
5
6  /* Use named colors */
7  input[type="radio"] {
8      accent-color: blue;
9  }
10
11 /* Use HSL color space */
12 progress {
13     accent-color: hsl(200, 100%, 50%);
14 }
15
16 /* Use RGB */
17 input[type="range"] {
18     accent-color: rgb(76, 175, 80);
19 }
20
21 /* Inherit from parent */
22 input {
23     accent-color: inherit;
24 }
25
26 /* Use auto (respects system preferences) */
27 input {
28     accent-color: auto;
29 }
```

Listing 2.1: Basic accent-color Syntax

2.2 Color Value Types

```
1 <style>
2   /* Hex colors */
3   .checkbox-hex {
4     accent-color: #667eea;
5   }
6
7   /* RGB/RGBA */
8   .checkbox-rgb {
9     accent-color: rgb(102, 126, 234);
10  }
11
12  .checkbox-rgba {
13    accent-color: rgba(102, 126, 234, 0.8);
14  }
15
16  /* HSL */
17  .checkbox-hsl {
18    accent-color: hsl(250, 100%, 66%);
19  }
20
21  /* HSL with alpha */
22  .checkbox-hsla {
23    accent-color: hsla(250, 100%, 66%, 0.8);
24  }
25
26  /* Named colors */
27  .checkbox-named {
28    accent-color: mediumvioletred;
29  }
30
31  /* CSS custom properties */
32  :root {
33    --brand-color: #ff6b6b;
34  }
35
36  .checkbox-custom {
37    accent-color: var(--brand-color);
38  }
39
40  /* Color mix function (modern browsers) */
41  .checkbox-mix {
42    accent-color: color-mix(in srgb, blue 80%, red 20%);
43  }
44 </style>
45
46 <input type="checkbox" class="checkbox-hex" />
47 <input type="checkbox" class="checkbox-rgb" />
48 <input type="checkbox" class="checkbox-hsl" />
49 <input type="checkbox" class="checkbox-named" />
50 <input type="checkbox" class="checkbox-custom" />
```

Listing 2.2: Different Color Value Options

2.3 Inheritance Behavior

```
1 <style>
2   /* Set on parent: applies to all children */
3   .form-group {
4     accent-color: #2196f3;
5   }
6
7   .form-group input,
8   .form-group progress {
9     /* Inherit the accent-color from parent */
10  }
11
12  /* Override in child */
13  .form-group input.special {
14    accent-color: #ff9800;
15  }
16
17  /* Use inherit keyword explicitly */
18  .form-group textarea {
19    accent-color: inherit;
20  }
21 </style>
22
23 <div class="form-group">
24   <label>
25     <input type="checkbox" /> Remember me
26   </label>
27   <label>
28     <input type="checkbox" class="special" /> Special option
29   </label>
30   <div>
31     <progress value="70" max="100"></progress>
32   </div>
33 </div>
```

Listing 2.3: Accent-Color Inheritance

Chapter 3

Practical Applications

3.1 Form Styling with Brand Colors

```
1 <style>
2   :root {
3     --primary-color: #667eea;
4     --primary-light: #f3f4ff;
5     --border-color: #e0e0e0;
6   }
7
8   .custom-form {
9     background-color: white;
10    padding: 30px;
11    border-radius: 8px;
12    max-width: 500px;
13    box-shadow: 0 2px 12px rgba(0, 0, 0, 0.1);
14  }
15
16  .custom-form h2 {
17    color: var(--primary-color);
18    margin-top: 0;
19  }
20
21  .form-group {
22    margin-bottom: 20px;
23    accent-color: var(--primary-color);
24  }
25
26  .form-group label {
27    display: block;
28    margin-bottom: 8px;
29    font-weight: 500;
30    color: #333;
31  }
32
33  .form-group input[type="text"],
34  .form-group input[type="email"],
35  .form-group input[type="password"] {
36    width: 100%;
```

```
37     padding: 10px 12px;
38     border: 1px solid var(--border-color);
39     border-radius: 4px;
40     font-family: inherit;
41     font-size: 14px;
42 }
43
44 .form-group input[type="text"]:focus,
45 .form-group input[type="email"]:focus,
46 .form-group input[type="password"]:focus {
47     outline: none;
48     border-color: var(--primary-color);
49     box-shadow: 0 0 0 3px var(--primary-light);
50 }
51
52 .checkbox-group {
53     background-color: var(--primary-light);
54     padding: 15px;
55     border-radius: 4px;
56     margin-bottom: 20px;
57 }
58
59 .checkbox-group label {
60     display: flex;
61     align-items: center;
62     margin-bottom: 10px;
63 }
64
65 .checkbox-group label:last-child {
66     margin-bottom: 0;
67 }
68
69 .checkbox-group input[type="checkbox"] {
70     margin-right: 10px;
71     width: 18px;
72     height: 18px;
73     cursor: pointer;
74 }
75
76 .button-group {
77     display: flex;
78     gap: 10px;
79 }
80
81 .button-group button {
82     flex: 1;
83     padding: 12px;
84     border: none;
85     border-radius: 4px;
86     font-weight: 600;
87     cursor: pointer;
88     transition: all 0.2s ease;
89 }
90
```

```

91     .button-group button.submit {
92         background-color: var(--primary-color);
93         color: white;
94     }
95
96     .button-group button.submit:hover {
97         background-color: #5568d3;
98     }
99
100    .button-group button.reset {
101        background-color: #f5f5f5;
102        color: #333;
103    }
104
105    .button-group button.reset:hover {
106        background-color: #efefef;
107    }
108 </style>
109
110 <form class="custom-form">
111     <h2>Sign Up</h2>
112
113     <div class="form-group">
114         <label for="username">Username</label>
115         <input type="text" id="username" placeholder="Choose a username" />
116     </div>
117
118     <div class="form-group">
119         <label for="email">Email</label>
120         <input type="email" id="email" placeholder="your@email.com" />
121     </div>
122
123     <div class="form-group">
124         <label for="password">Password</label>
125         <input type="password" id="password" placeholder="....." />
126     </div>
127
128     <div class="form-group">
129         <label for="confirm">Confirm Password</label>
130         <input type="password" id="confirm" placeholder="....." />
131     </div>
132
133     <div class="checkbox-group">
134         <label>
135             <input type="checkbox" checked /> I agree to terms
136         </label>
137         <label>
138             <input type="checkbox" checked /> I want email updates
139         </label>
140         <label>
141             <input type="checkbox" /> I am 18+ years old
142         </label>
143     </div>
144

```

```
145 <div class="button-group">
146   <button type="reset" class="reset">Reset</button>
147   <button type="submit" class="submit">Create Account</button>
148 </div>
149 </form>
```

Listing 3.1: Complete Form with accent-color

3.2 Checkbox Styling

```
1 <style>
2   .checkbox-showcase {
3     padding: 20px;
4     background-color: #f5f5f5;
5   }
6
7   .checkbox-group-demo {
8     margin-bottom: 30px;
9     background-color: white;
10    padding: 20px;
11    border-radius: 8px;
12  }
13
14  .checkbox-group-demo h3 {
15    margin-top: 0;
16    color: #333;
17    font-size: 14px;
18    text-transform: uppercase;
19    letter-spacing: 0.5px;
20  }
21
22  .checkbox-item {
23    display: flex;
24    align-items: center;
25    margin-bottom: 12px;
26  }
27
28  .checkbox-item:last-child {
29    margin-bottom: 0;
30  }
31
32  .checkbox-item input[type="checkbox"] {
33    margin-right: 12px;
34    width: 20px;
35    height: 20px;
36    cursor: pointer;
37  }
38
39  .checkbox-item label {
40    cursor: pointer;
41    user-select: none;
42    color: #555;
```

```

43     }
44
45     /* Brand colors */
46     .group-purple .checkbox-item input[type="checkbox"] {
47         accent-color: #9c27b0;
48     }
49
50     .group-blue .checkbox-item input[type="checkbox"] {
51         accent-color: #2196f3;
52     }
53
54     .group-green .checkbox-item input[type="checkbox"] {
55         accent-color: #4caf50;
56     }
57
58     .group-orange .checkbox-item input[type="checkbox"] {
59         accent-color: #ff9800;
60     }
61
62     .group-red .checkbox-item input[type="checkbox"] {
63         accent-color: #f44336;
64     }
65 </style>
66
67 <div class="checkbox-showcase">
68     <div class="checkbox-group-demo group-purple">
69         <h3>Purple Theme</h3>
70         <div class="checkbox-item">
71             <input type="checkbox" id="p1" />
72             <label for="p1">Option A</label>
73         </div>
74         <div class="checkbox-item">
75             <input type="checkbox" id="p2" checked />
76             <label for="p2">Option B (checked)</label>
77         </div>
78         <div class="checkbox-item">
79             <input type="checkbox" id="p3" />
80             <label for="p3">Option C</label>
81         </div>
82     </div>
83
84     <div class="checkbox-group-demo group-blue">
85         <h3>Blue Theme</h3>
86         <div class="checkbox-item">
87             <input type="checkbox" id="b1" />
88             <label for="b1">Feature X</label>
89         </div>
90         <div class="checkbox-item">
91             <input type="checkbox" id="b2" checked />
92             <label for="b2">Feature Y (checked)</label>
93         </div>
94     </div>
95
96     <div class="checkbox-group-demo group-green">

```

```
97     <h3>Green Theme</h3>
98     <div class="checkbox-item">
99         <input type="checkbox" id="g1" checked />
100         <label for="g1">Enabled</label>
101     </div>
102 </div>
103 </div>
```

Listing 3.2: Branded Checkbox Inputs

3.3 Radio Button Styling

```
1 <style>
2   .radio-group {
3     padding: 20px;
4     background-color: white;
5     border-radius: 8px;
6     box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1);
7     max-width: 400px;
8   }
9
10  .radio-group h3 {
11    margin-top: 0;
12    color: #333;
13  }
14
15  .radio-item {
16    display: flex;
17    align-items: center;
18    margin-bottom: 16px;
19    padding: 12px;
20    background-color: #f9f9f9;
21    border-radius: 4px;
22    transition: background-color 0.2s;
23  }
24
25  .radio-item:hover {
26    background-color: #f5f5f5;
27  }
28
29  .radio-item input[type="radio"] {
30    margin-right: 12px;
31    width: 18px;
32    height: 18px;
33    cursor: pointer;
34    accent-color: #667eea;
35  }
36
37  .radio-item label {
38    flex: 1;
39    cursor: pointer;
40    user-select: none;
```

```
41     }
42
43     .radio-item input[type="radio"]:checked + label {
44         font-weight: 600;
45         color: #667eea;
46     }
47
48     .radio-description {
49         display: block;
50         font-size: 12px;
51         color: #999;
52         margin-left: 30px;
53         margin-top: 4px;
54     }
55 </style>
56
57 <div class="radio-group">
58     <h3>Select a subscription plan</h3>
59
60     <div class="radio-item">
61         <input type="radio" id="free" name="plan" value="free" checked />
62         <div>
63             <label for="free">Free Plan</label>
64             <span class="radio-description">Limited features, perfect for
              testing</span>
65         </div>
66     </div>
67
68     <div class="radio-item">
69         <input type="radio" id="pro" name="plan" value="pro" />
70         <div>
71             <label for="pro">Pro Plan</label>
72             <span class="radio-description">Advanced features, billed monthly
              </span>
73         </div>
74     </div>
75
76     <div class="radio-item">
77         <input type="radio" id="enterprise" name="plan" value="enterprise" />
78         <div>
79             <label for="enterprise">Enterprise Plan</label>
80             <span class="radio-description">Custom features, dedicated support
              </span>
81         </div>
82     </div>
83 </div>
```

Listing 3.3: Branded Radio Buttons

Chapter 4

Range and Progress Elements

4.1 Range Input Styling

```
1 <style>
2   .range-demo {
3     padding: 30px;
4     background-color: white;
5     border-radius: 8px;
6     max-width: 400px;
7   }
8
9   .range-demo h3 {
10     margin-top: 0;
11   }
12
13   .range-group {
14     margin-bottom: 30px;
15   }
16
17   .range-group label {
18     display: block;
19     margin-bottom: 10px;
20     font-weight: 500;
21     color: #333;
22   }
23
24   .range-group input[type="range"] {
25     width: 100%;
26     height: 6px;
27     border-radius: 3px;
28     outline: none;
29     -webkit-appearance: none;
30     appearance: none;
31   }
32
33   .range-group input[type="range"]::-webkit-slider-thumb {
34     -webkit-appearance: none;
35     appearance: none;
36     width: 20px;
```



```
37     height: 20px;
38     border-radius: 50%;
39     background-color: #667eea;
40     cursor: pointer;
41 }
42
43 .range-group input[type="range"]::-moz-range-thumb {
44     width: 20px;
45     height: 20px;
46     border-radius: 50%;
47     background-color: #667eea;
48     cursor: pointer;
49     border: none;
50 }
51
52 /* Using accent-color (simpler approach) */
53 .range-modern {
54     accent-color: #667eea;
55 }
56
57 .range-value {
58     display: inline-block;
59     margin-left: 10px;
60     font-weight: 600;
61     color: #667eea;
62     min-width: 30px;
63 }
64 </style>
65
66 <div class="range-demo">
67     <h3>Range Slider Examples</h3>
68
69     <div class="range-group">
70         <label>Volume Control (Traditional)</label>
71         <input type="range" min="0" max="100" value="50" />
72     </div>
73
74     <div class="range-group">
75         <label>
76             Brightness Adjustment
77             <span class="range-value" id="brightness-value">60</span>%
78         </label>
79         <input type="range" class="range-modern" min="0" max="100" value="60"
80             id="brightness" />
81     </div>
82
83     <div class="range-group">
84         <label>
85             Speed
86             <span class="range-value" id="speed-value">50</span> km/h
87         </label>
88         <input type="range" class="range-modern" min="0" max="200" value="50"
89             id="speed" />
90     </div>
```

```
89     <div class="range-group">
90         <label>Opacity</label>
91         <input type="range" class="range-modern" min="0" max="100" value="100"
92             />
93     </div>
94 </div>
95
96 <script>
97     document.getElementById('brightness').addEventListener('input', function()
98     {
99         document.getElementById('brightness-value').textContent = this.value;
100     });
101
102     document.getElementById('speed').addEventListener('input', function() {
103         document.getElementById('speed-value').textContent = this.value;
104     });
105 </script>
```

Listing 4.1: Styled Range Slider

4.2 Progress Bar Styling

```
1 <style>
2     .progress-demo {
3         padding: 30px;
4         background-color: #f5f5f5;
5         border-radius: 8px;
6     }
7
8     .progress-section {
9         margin-bottom: 30px;
10        background-color: white;
11        padding: 20px;
12        border-radius: 8px;
13    }
14
15    .progress-section h4 {
16        margin-top: 0;
17        color: #333;
18    }
19
20    .progress-item {
21        margin-bottom: 20px;
22    }
23
24    .progress-item:last-child {
25        margin-bottom: 0;
26    }
27
28    .progress-label {
29        display: flex;
```

```
30     justify-content: space-between;
31     margin-bottom: 8px;
32     font-size: 14px;
33     font-weight: 500;
34 }
35
36 .progress-label .label-name {
37     color: #333;
38 }
39
40 .progress-label .label-percent {
41     color: #999;
42 }
43
44 progress {
45     width: 100%;
46     height: 8px;
47     border-radius: 4px;
48     overflow: hidden;
49 }
50
51 /* Default colors */
52 progress.primary {
53     accent-color: #667eea;
54 }
55
56 progress.success {
57     accent-color: #4caf50;
58 }
59
60 progress.warning {
61     accent-color: #ff9800;
62 }
63
64 progress.danger {
65     accent-color: #f44336;
66 }
67
68 progress.info {
69     accent-color: #2196f3;
70 }
71 </style>
72
73 <div class="progress-demo">
74     <div class="progress-section">
75         <h4>Project Progress</h4>
76
77         <div class="progress-item">
78             <div class="progress-label">
79                 <span class="label-name">Design</span>
80                 <span class="label-percent">100%</span>
81             </div>
82             <progress class="success" value="100" max="100"></progress>
83         </div>
```

```
84
85     <div class="progress-item">
86         <div class="progress-label">
87             <span class="label-name">Development</span>
88             <span class="label-percent">75%</span>
89         </div>
90         <progress class="primary" value="75" max="100"></progress>
91     </div>
92
93     <div class="progress-item">
94         <div class="progress-label">
95             <span class="label-name">Testing</span>
96             <span class="label-percent">45%</span>
97         </div>
98         <progress class="warning" value="45" max="100"></progress>
99     </div>
100
101     <div class="progress-item">
102         <div class="progress-label">
103             <span class="label-name">Deployment</span>
104             <span class="label-percent">10%</span>
105         </div>
106         <progress class="danger" value="10" max="100"></progress>
107     </div>
108 </div>
109
110 <div class="progress-section">
111     <h4>Download Speeds</h4>
112
113     <div class="progress-item">
114         <div class="progress-label">
115             <span class="label-name">File 1: document.pdf</span>
116             <span class="label-percent">95%</span>
117         </div>
118         <progress class="info" value="95" max="100"></progress>
119     </div>
120
121     <div class="progress-item">
122         <div class="progress-label">
123             <span class="label-name">File 2: video.mp4</span>
124             <span class="label-percent">60%</span>
125         </div>
126         <progress class="info" value="60" max="100"></progress>
127     </div>
128 </div>
129 </div>
```

Listing 4.2: Styled Progress Elements

Chapter 5

Advanced Applications

5.1 Multi-Theme Systems

```
1 <style>
2   :root {
3     --accent-color: #667eea;
4   }
5
6   /* Light theme */
7   [data-theme="light"] {
8     --accent-color: #667eea;
9   }
10
11  /* Dark theme */
12  [data-theme="dark"] {
13    --accent-color: #a78bfa;
14  }
15
16  /* High contrast theme */
17  [data-theme="high-contrast"] {
18    --accent-color: #ffff00;
19  }
20
21  input,
22  progress,
23  meter {
24    accent-color: var(--accent-color);
25  }
26
27  .theme-switcher {
28    padding: 20px;
29    background-color: #f5f5f5;
30    border-radius: 8px;
31    margin-bottom: 20px;
32  }
33
34  .theme-button {
35    padding: 8px 16px;
36    margin-right: 8px;
```

```

37     border: 2px solid #ddd;
38     background-color: white;
39     border-radius: 4px;
40     cursor: pointer;
41     transition: all 0.2s;
42 }
43
44 .theme-button.active {
45     background-color: var(--accent-color);
46     color: white;
47     border-color: var(--accent-color);
48 }
49
50 .demo-content {
51     background-color: white;
52     padding: 20px;
53     border-radius: 8px;
54 }
55 </style>
56
57 <div class="theme-switcher">
58     <button class="theme-button active" onclick="setTheme('light')">Light</button>
59     <button class="theme-button" onclick="setTheme('dark')">Dark</button>
60     <button class="theme-button" onclick="setTheme('high-contrast')">High
        Contrast</button>
61 </div>
62
63 <div class="demo-content" data-theme="light">
64     <h3>Themed Form Elements</h3>
65     <label>
66         <input type="checkbox" checked /> Option 1
67     </label><br/>
68     <label>
69         <input type="radio" name="demo" checked /> Choice A
70     </label><br/>
71     <label>
72         <input type="range" value="50" />
73     </label><br/>
74     <progress value="70" max="100"></progress>
75 </div>
76
77 <script>
78     function setTheme(theme) {
79         document.querySelector('[data-theme]').setAttribute('data-theme',
            theme);
80         document.querySelectorAll('.theme-button').forEach(btn => {
81             btn.classList.remove('active');
82         });
83         event.target.classList.add('active');
84     }
85 </script>

```

Listing 5.1: Theme-Aware accent-color

5.2 Dynamic Accent Colors

```
1 <style>
2   .color-picker-demo {
3     padding: 20px;
4     background-color: white;
5     border-radius: 8px;
6     max-width: 500px;
7   }
8
9   .color-input-group {
10    margin-bottom: 20px;
11    display: flex;
12    gap: 10px;
13    align-items: center;
14  }
15
16  .color-picker-input {
17    width: 60px;
18    height: 40px;
19    border: none;
20    border-radius: 4px;
21    cursor: pointer;
22  }
23
24  .controls-section {
25    background-color: #f9f9f9;
26    padding: 15px;
27    border-radius: 4px;
28    margin-bottom: 20px;
29  }
30
31  .preview-grid {
32    display: grid;
33    grid-template-columns: repeat(2, 1fr);
34    gap: 20px;
35  }
36
37  .preview-item {
38    padding: 15px;
39    border: 1px solid #ddd;
40    border-radius: 4px;
41  }
42
43  .preview-item h4 {
44    margin-top: 0;
45    font-size: 12px;
46    color: #999;
47    text-transform: uppercase;
48  }
49
50  input[type="checkbox"],
51  input[type="radio"],
```

```
52     input[type="range"] {
53         accent-color: var(--preview-color, #667eea);
54     }
55
56     progress {
57         accent-color: var(--preview-color, #667eea);
58         width: 100%;
59     }
60 </style>
61
62 <div class="color-picker-demo">
63     <div class="color-input-group">
64         <label>Choose accent color:</label>
65         <input type="color" id="colorPicker" value="#667eea"
66             onchange="updateAccentColor(this.value)" />
67         <span id="colorValue">#667eea</span>
68     </div>
69
70     <div class="preview-grid">
71         <div class="preview-item">
72             <h4>Checkbox</h4>
73             <label>
74                 <input type="checkbox" checked /> Option
75             </label>
76         </div>
77
78         <div class="preview-item">
79             <h4>Radio</h4>
80             <label>
81                 <input type="radio" checked /> Choice
82             </label>
83         </div>
84
85         <div class="preview-item">
86             <h4>Range</h4>
87             <input type="range" value="50" />
88         </div>
89
90         <div class="preview-item">
91             <h4>Progress</h4>
92             <progress value="75" max="100"></progress>
93         </div>
94     </div>
95 </div>
96
97 <script>
98     function updateAccentColor(color) {
99         document.documentElement.style.setProperty('--preview-color', color);
100         document.getElementById('colorValue').textContent = color;
101     }
102 </script>
```

Listing 5.2: Programmatic Color Changes

Chapter 6

Fallbacks and Browser Support

6.1 Feature Detection

```
1 <style>
2   /* Modern browsers: use accent-color */
3   @supports (accent-color: red) {
4     input[type="checkbox"],
5     input[type="radio"],
6     progress {
7       accent-color: #667eea;
8     }
9   }
10
11  /* Fallback for older browsers */
12  @supports not (accent-color: red) {
13    /* Use alternative styling */
14    input[type="checkbox"] {
15      /* Styling workaround */
16    }
17  }
18 </style>
```

Listing 6.1: Detecting accent-color Support

6.2 Progressive Enhancement

```
1 <style>
2   /* Step 1: Base styling works everywhere */
3   input[type="checkbox"],
4   input[type="radio"],
5   progress {
6     /* Basic styling that works in all browsers */
7   }
8
9   /* Step 2: Enhanced with accent-color */
10  input[type="checkbox"],
11  input[type="radio"],
```

```
12   progress {
13       accent-color: #667eea;
14   }
15
16   /* Step 3: Modern color features */
17   @supports (accent-color: color-mix(in srgb, blue, red)) {
18       .adaptive-accent {
19           accent-color: color-mix(in srgb,
20               var(--primary-color) 70%,
21               var(--secondary-color) 30%);
22       }
23   }
24 </style>
```

Listing 6.2: Graceful Degradation

6.3 Browser Compatibility Table

Browser	Version	Support	Checkboxes	Radios	Range
Chrome	93+	Full	Yes	Yes	Partial
Firefox	92+	Full	Yes	Yes	Yes
Safari	15.4+	Full	Yes	Yes	Yes
Edge	93+	Full	Yes	Yes	Partial
IE 11	All	No	No	No	No

Chapter 7

Color Input with accent-color

7.1 Color Input Styling

```
1 <style>
2   .color-input-demo {
3     padding: 20px;
4     background-color: white;
5     border-radius: 8px;
6     max-width: 400px;
7   }
8
9   .color-input-group {
10     margin-bottom: 20px;
11   }
12
13   .color-input-group label {
14     display: block;
15     margin-bottom: 8px;
16     font-weight: 500;
17     color: #333;
18   }
19
20   .color-input-group input[type="color"] {
21     width: 100%;
22     height: 60px;
23     border: 2px solid #ddd;
24     border-radius: 8px;
25     cursor: pointer;
26     accent-color: #667eea;
27   }
28
29   .color-input-group input[type="color"]:focus {
30     outline: none;
31     border-color: #667eea;
32   }
33
34   .color-palette {
35     display: grid;
36     grid-template-columns: repeat(4, 1fr);
```

```
37     gap: 10px;
38     margin-bottom: 20px;
39 }
40
41 .color-swatch {
42     width: 100%;
43     aspect-ratio: 1;
44     border-radius: 8px;
45     cursor: pointer;
46     border: 2px solid transparent;
47     transition: all 0.2s;
48 }
49
50 .color-swatch:hover {
51     border-color: #333;
52     transform: scale(1.05);
53 }
54 </style>
55
56 <div class="color-input-demo">
57     <div class="color-input-group">
58         <label for="primary">Select Primary Color</label>
59         <input type="color" id="primary" value="#667eea" />
60     </div>
61
62     <div class="color-palette">
63         <div class="color-swatch" style="background-color: #FF6B6B;"></div>
64         <div class="color-swatch" style="background-color: #4ECDC4;"></div>
65         <div class="color-swatch" style="background-color: #FFE66D;"></div>
66         <div class="color-swatch" style="background-color: #95E1D3;"></div>
67     </div>
68 </div>
```

Listing 7.1: Styled Color Input

Chapter 8

Real-World Examples

8.1 Settings Panel

```
1 <style>
2   .settings-panel {
3     max-width: 600px;
4     background-color: white;
5     border-radius: 8px;
6     box-shadow: 0 2px 12px rgba(0, 0, 0, 0.1);
7     overflow: hidden;
8   }
9
10  .settings-header {
11    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
12    color: white;
13    padding: 24px;
14  }
15
16  .settings-header h2 {
17    margin: 0;
18    font-size: 24px;
19  }
20
21  .settings-body {
22    padding: 24px;
23  }
24
25  .setting-group {
26    margin-bottom: 28px;
27    padding-bottom: 28px;
28    border-bottom: 1px solid #eee;
29  }
30
31  .setting-group:last-child {
32    margin-bottom: 0;
33    padding-bottom: 0;
34    border-bottom: none;
35  }
36
```

```
37 .setting-title {
38     font-weight: 600;
39     color: #333;
40     margin-bottom: 12px;
41 }
42
43 .setting-description {
44     font-size: 13px;
45     color: #999;
46     margin-bottom: 12px;
47 }
48
49 .setting-options {
50     display: flex;
51     flex-direction: column;
52     gap: 10px;
53 }
54
55 .setting-option {
56     display: flex;
57     align-items: center;
58 }
59
60 .setting-option input {
61     margin-right: 12px;
62     accent-color: #667eea;
63 }
64
65 .setting-option label {
66     cursor: pointer;
67     user-select: none;
68 }
69
70 .range-setting {
71     display: flex;
72     align-items: center;
73     gap: 12px;
74 }
75
76 .range-setting input[type="range"] {
77     flex: 1;
78     accent-color: #667eea;
79 }
80
81 .range-value {
82     min-width: 40px;
83     text-align: right;
84     font-weight: 600;
85 }
86
87 .button-group {
88     display: flex;
89     gap: 12px;
90     justify-content: flex-end;
```

```
91     margin-top: 24px;
92     padding-top: 24px;
93     border-top: 1px solid #eee;
94 }
95
96 .button-group button {
97     padding: 10px 20px;
98     border: 1px solid #ddd;
99     background-color: white;
100     border-radius: 4px;
101     cursor: pointer;
102     font-weight: 500;
103     transition: all 0.2s;
104 }
105
106 .button-group button.save {
107     background-color: #667eea;
108     color: white;
109     border-color: #667eea;
110 }
111
112 .button-group button.save:hover {
113     background-color: #5568d3;
114 }
115 </style>
116
117 <div class="settings-panel">
118     <div class="settings-header">
119         <h2>Settings</h2>
120     </div>
121     <div class="settings-body">
122         <div class="setting-group">
123             <div class="setting-title">Notifications</div>
124             <div class="setting-description">Manage how you receive
125                 notifications</div>
126             <div class="setting-options">
127                 <div class="setting-option">
128                     <input type="checkbox" id="email-notif" checked />
129                     <label for="email-notif">Email notifications</label>
130                 </div>
131                 <div class="setting-option">
132                     <input type="checkbox" id="push-notif" checked />
133                     <label for="push-notif">Push notifications</label>
134                 </div>
135                 <div class="setting-option">
136                     <input type="checkbox" id="sms-notif" />
137                     <label for="sms-notif">SMS notifications</label>
138                 </div>
139             </div>
140         </div>
141         <div class="setting-group">
142             <div class="setting-title">Privacy</div>
```

```

143     <div class="setting-description">Control your privacy settings</div>
144     <div class="setting-options">
145         <div class="setting-option">
146             <input type="checkbox" id="profile-private" />
147             <label for="profile-private">Make profile private</label>
148         </div>
149         <div class="setting-option">
150             <input type="checkbox" id="hide-email" checked />
151             <label for="hide-email">Hide email address</label>
152         </div>
153         <div class="setting-option">
154             <input type="checkbox" id="analytics" checked />
155             <label for="analytics">Allow analytics tracking</label>
156         </div>
157     </div>
158 </div>
159
160 <div class="setting-group">
161     <div class="setting-title">Preferences</div>
162     <div class="setting-description">Customize your experience</div>
163     <div class="range-setting">
164         <label>Font Size:</label>
165         <input type="range" min="12" max="20" value="14" />
166         <span class="range-value" id="font-size">14px</span>
167     </div>
168 </div>
169
170 <div class="button-group">
171     <button>Cancel</button>
172     <button class="save">Save Changes</button>
173 </div>
174 </div>
175 </div>

```

Listing 8.1: Complete Settings Component

8.2 Survey Form

```

1 <style>
2   .survey-form {
3     max-width: 500px;
4     background-color: white;
5     border-radius: 8px;
6     padding: 30px;
7     box-shadow: 0 2px 12px rgba(0, 0, 0, 0.1);
8   }
9
10  .survey-form h1 {
11    margin-top: 0;
12    color: #667eea;
13    text-align: center;

```



```
14 }
15
16 .question-block {
17     margin-bottom: 28px;
18 }
19
20 .question-title {
21     font-weight: 600;
22     color: #333;
23     margin-bottom: 12px;
24     display: flex;
25     align-items: center;
26 }
27
28 .question-number {
29     display: inline-block;
30     width: 24px;
31     height: 24px;
32     background-color: #667eea;
33     color: white;
34     border-radius: 50%;
35     text-align: center;
36     line-height: 24px;
37     font-size: 12px;
38     margin-right: 8px;
39 }
40
41 .question-options {
42     display: flex;
43     flex-direction: column;
44     gap: 10px;
45     margin-left: 32px;
46 }
47
48 .option-item {
49     display: flex;
50     align-items: center;
51 }
52
53 .option-item input {
54     margin-right: 10px;
55     accent-color: #667eea;
56 }
57
58 .progress-section {
59     margin-bottom: 24px;
60     padding-bottom: 24px;
61     border-bottom: 2px solid #f0f0f0;
62 }
63
64 .progress-info {
65     display: flex;
66     justify-content: space-between;
67     margin-bottom: 8px;
```

```

68     font-size: 12px;
69 }
70
71 progress {
72     width: 100%;
73     height: 6px;
74     accent-color: #667eea;
75 }
76
77 .submit-button {
78     width: 100%;
79     padding: 12px;
80     background-color: #667eea;
81     color: white;
82     border: none;
83     border-radius: 4px;
84     font-size: 16px;
85     font-weight: 600;
86     cursor: pointer;
87     margin-top: 24px;
88 }
89
90 .submit-button:hover {
91     background-color: #5568d3;
92 }
93 </style>
94
95 <form class="survey-form">
96     <h1>Customer Satisfaction Survey</h1>
97
98     <div class="progress-section">
99         <div class="progress-info">
100             <span>Progress</span>
101             <span>2 of 5</span>
102         </div>
103         <progress value="2" max="5"></progress>
104     </div>
105
106     <div class="question-block">
107         <div class="question-title">
108             <span class="question-number">1</span>
109             How satisfied are you with our service?
110         </div>
111         <div class="question-options">
112             <div class="option-item">
113                 <input type="radio" id="q1a" name="q1" />
114                 <label for="q1a">Very Satisfied</label>
115             </div>
116             <div class="option-item">
117                 <input type="radio" id="q1b" name="q1" checked />
118                 <label for="q1b">Satisfied</label>
119             </div>
120             <div class="option-item">
121                 <input type="radio" id="q1c" name="q1" />

```

```
122         <label for="q1c">Neutral</label>
123     </div>
124     <div class="option-item">
125         <input type="radio" id="q1d" name="q1" />
126         <label for="q1d">Dissatisfied</label>
127     </div>
128 </div>
129 </div>
130
131 <div class="question-block">
132     <div class="question-title">
133         <span class="question-number">2</span>
134         Would you recommend us to others?
135     </div>
136     <div class="question-options">
137         <div class="option-item">
138             <input type="checkbox" id="q2a" checked />
139             <label for="q2a">Yes, definitely</label>
140         </div>
141         <div class="option-item">
142             <input type="checkbox" id="q2b" checked />
143             <label for="q2b">Maybe</label>
144         </div>
145         <div class="option-item">
146             <input type="checkbox" id="q2c" />
147             <label for="q2c">No</label>
148         </div>
149     </div>
150 </div>
151
152     <button type="submit" class="submit-button">Submit Survey</button>
153 </form>
```

Listing 8.2: Survey with Themed Controls

Chapter 9

Quick Reference

9.1 Complete Property Reference

```
1  /* Syntax */
2  accent-color: <color>;
3  accent-color: auto;
4  accent-color: inherit;
5  accent-color: initial;
6  accent-color: revert;
7  accent-color: unset;
8
9  /* Color values */
10 accent-color: red;
11 accent-color: #FF0000;
12 accent-color: rgb(255, 0, 0);
13 accent-color: rgba(255, 0, 0, 0.5);
14 accent-color: hsl(0, 100%, 50%);
15 accent-color: hsla(0, 100%, 50%, 0.5);
16
17 /* CSS variables */
18 accent-color: var(--brand-color);
19 accent-color: var(--brand-color, blue);
20
21 /* Applying to elements */
22 input[type="checkbox"] { accent-color: blue; }
23 input[type="radio"] { accent-color: green; }
24 input[type="range"] { accent-color: orange; }
25 progress { accent-color: purple; }
26 meter { accent-color: teal; }
```

Listing 9.1: accent-color Property Cheat Sheet

9.2 Best Practices

1. Use CSS custom properties for theming consistency
2. Always test on multiple browsers before deployment
3. Provide fallback colors for older browsers

4. Ensure sufficient color contrast for accessibility
5. Use `@supports` to detect feature support
6. Consider prefers-color-scheme for dark mode
7. Don't rely on accent-color for critical UX features
8. Test with color blindness simulation tools
9. Use semantic color meanings (green = success, red = error)
10. Document theme colors in design systems

9.3 Supported Elements Summary

- `<input type="checkbox">`
- `<input type="radio">`
- `<input type="range">` (partial)
- `<progress>`
- `<meter>`
- `<input type="color">`
- Other input types
- Text content

Chapter 10

Conclusion

10.1 Summary

The CSS `accent-color` property provides a simple, elegant solution for theming native form elements. With universal modern browser support and straightforward syntax, it's now the recommended approach for customizing checkbox, radio button, range, and progress element colors without complex CSS hacks or custom components.

10.2 Key Advantages

- Simple one-line color customization
- Works on native HTML elements
- Cascades through inheritance
- Respects system dark mode preferences
- Zero performance impact
- Cross-browser consistency
- Accessibility-friendly
- Perfect for theme systems

10.3 Related CSS Features

Explore these complementary CSS properties:

- `color` - Text color
- `background-color` - Background color
- `caret-color` - Text cursor color
- `outline` - Focus outline styling
- `border` - Element borders